

T3 matched BSM $\Delta\mathcal{L}_{\text{EFT}}$ terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from matched BSM EFT terms. Empty cells denote no matched term with that field combination.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$ $-\frac{1}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$ $-\frac{1}{4}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$+C_{\text{H2}}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_{\text{H1}}^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{1}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}\lambda_{\text{H1}}^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{3\hbar H_{a_1}H_{a_2}\lambda_{\text{H2}}^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{3}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}\lambda_{\text{H2}}^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\hbar\lambda_{\text{T3}}H_{a_1}H_{a_2}\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{\hbar\lambda_{\text{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\lambda_{\text{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\lambda_{\text{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{2(M_{S_1}^2-M_{S_2}^2)}$	$-\frac{2\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon}$ $-\frac{1}{2}\hbar \log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$ $-\frac{1}{6}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$	$-\frac{\hbar \log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W}_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}\ell_{e_4,\tau_2}^TC P_\ell \ell_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\tau_1}\ell_{a_1,a_2}\ell_{a_3,a_4}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{\hbar \log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W}_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}\ell_{e_4,\tau_2}^TC P_\ell \ell_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau_1}\ell_{a_1,a_2}\ell_{a_3,a_4}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{\hbar \log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W}_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}C P_h \ell_{a_4,\tau_2}^TM_{S_1}^2W_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}\ell_{a_1,a_3}\ell_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{\hbar \log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W}_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}\ell_{e_4,\tau_2}^TC P_\ell \ell_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_1}^2M_{S_2}^2\lambda_{\tau_1}\ell_{a_1,a_2}\ell_{a_3,a_4}M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{\hbar \log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W}_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}\ell_{e_4,\tau_2}^TC P_\ell \ell_{a_2,\tau_1}H_{a_1}H_{a_2}M_{S_2}^2M_{S_1}^2\lambda_{\tau_1}\ell_{a_1,a_2}\ell_{a_3,a_4}M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{\hbar \log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W}_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}C P_h \overline{P}_{a_4,\tau_2}^TM_{S_1}^2M_{S_2}^2W_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}\ell_{a_1,a_3}\ell_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{\hbar \log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W}_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}C P_h \overline{P}_{a_4,\tau_2}^TM_{S_1}^2M_{S_2}^2W_{\tau_1,\tau_1}\overline{W}_{\tau_2,\tau_1}\ell_{a_1,a_3}\ell_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$
	T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\hbar B_{\mu_1\mu_2}^2}{12\epsilon}$ $-\frac{1}{24}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)B_{\mu_1\mu_2}^2$ $-\frac{1}{24}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$+C_{\text{H2}}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_{\text{H1}}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}\lambda_{\text{H1}}^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\hbar H_{a_1}H_{a_2}\lambda_{\text{H2}}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}\lambda_{\text{H2}}^2H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\hbar\lambda_{\text{T3}}H_{a_1}H_{a_2}\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{\hbar\lambda_{\text{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $+\frac{\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\lambda_{\text{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $-\frac{\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\lambda_{\text{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\text{T3}}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$	$-\frac{\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon}$ $-\frac{1}{24}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$ $-\frac{1}{24}\hbar \log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-C	-1	2	-	$\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$ \begin{aligned} & +C_{H2}H_{a_1}H_{a_1}^\dagger \\ & -H_{a_1}\mu^2H_{a_1}^\dagger \\ & +\frac{\hbar H_{a_1}H_{a_2}\lambda_{H1}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\frac{H_{a_1}H_{a_2}\lambda_{H1}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{\hbar H_{a_1}H_{a_2}\lambda_{H2}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\frac{H_{a_1}H_{a_2}\lambda_{H2}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2} \\ & +\frac{\hbar\overline{\lambda_{T5}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T5}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2} \\ & +\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2} \\ & -\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2} \\ & -\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S1}^2-M_{S2}^2} \end{aligned} $	$ \begin{aligned} & -\frac{13\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{12\epsilon} \\ & -\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{24}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \end{aligned} $	$ \begin{aligned} & -\frac{\hbar\log\left(\frac{\mu\ell^2}{M_{S1}^2}\right)\overline{W_{T1,T1}W_{T2,T2}}\overline{\ell_{a_1,T1}^T}CP_{\ell_{a_2,T2}}H_{a_2}H_{a_3}M_{S1}^2\lambda_{T1}\ell_{a_1,T2}\ell_{a_3,T4}M_F^2}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\ell^2}{M_{S1}^2}\right)\overline{W_{T1,T1}W_{T2,T2}}\overline{\ell_{a_1,T1}^T}CP_{\ell_{a_2,T2}}H_{a_2}H_{a_3}M_{S1}^2\lambda_{T1}\ell_{a_1,T2}\ell_{a_3,T4}M_F^2}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\ell^2}{M_F^2}\right)\overline{\lambda_{T1}\ell_{a_2,T2}}CP_{\ell_{a_3,T3}}\ell_{a_1,T1}^TM_{S1}^2W_{T1,T1}W_{T2,T2}\ell_{T1}\ell_{a_2,T3}\ell_{a_3,T4}H_{a_1}^{\dagger}H_{a_2}^{\dagger}M_F^2}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\ell^2}{M_F^2}\right)\overline{\lambda_{T1}\ell_{a_2,T2}}CP_{\ell_{a_3,T3}}\ell_{a_1,T1}^TM_{S1}^2W_{T1,T1}W_{T2,T2}\ell_{T1}\ell_{a_2,T3}\ell_{a_3,T4}H_{a_1}^{\dagger}H_{a_2}^{\dagger}M_F^2}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & -\frac{\hbar\log\left(\frac{\mu\ell^2}{M_{S1}^2}\right)\overline{W_{T1,T1}W_{T2,T2}}\ell_{T1}\ell_{a_4,T2}^TC P_{\ell_{a_2,T2}}H_{a_2}H_{a_3}M_{S1}^2\lambda_{T1}\ell_{a_1,T2}\ell_{a_3,T4}M_F}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\ell^2}{M_{S1}^2}\right)\overline{W_{T1,T1}W_{T2,T2}}\ell_{T1}\ell_{a_4,T2}^TC P_{\ell_{a_2,T2}}H_{a_2}H_{a_3}M_{S1}^2\lambda_{T1}\ell_{a_1,T2}\ell_{a_3,T4}M_F}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & -\frac{\hbar\log\left(\frac{\mu\ell^2}{M_F^2}\right)\overline{\lambda_{T1}\ell_{a_2,T2}}CP_{\ell_{a_3,T3}}\ell_{a_1,T1}^TM_{S1}^2M_{S2}^2W_{T1,T1}W_{T2,T2}\ell_{T1}\ell_{a_2,T3}\ell_{a_3,T4}H_{a_1}^{\dagger}H_{a_2}^{\dagger}M_F}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\ell^2}{M_F^2}\right)\overline{\lambda_{T1}\ell_{a_2,T2}}CP_{\ell_{a_3,T3}}\ell_{a_1,T1}^TM_{S1}^2M_{S2}^2W_{T1,T1}W_{T2,T2}\ell_{T1}\ell_{a_2,T3}\ell_{a_3,T4}H_{a_1}^{\dagger}H_{a_2}^{\dagger}M_F}{4\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \end{aligned} $
									$ \begin{aligned} & +C_{H2}H_{a_1}H_{a_1}^\dagger \\ & -H_{a_1}\mu^2H_{a_1}^\dagger \\ & +\frac{3\hbar H_{a_1}H_{a_2}\lambda_{H1}^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\frac{H_{a_1}H_{a_2}\lambda_{H1}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{\hbar H_{a_1}H_{a_2}\lambda_{H2}^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S2}^2}\right)\frac{H_{a_1}H_{a_2}\lambda_{H2}^2H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{\hbar\overline{\lambda_{T5}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T5}H_{a_1}^\dagger H_{a_2}^\dagger}{2\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{2\left(M_{S1}^2-M_{S2}^2\right)} \\ & -\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{2\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{2\left(M_{S1}^2-M_{S2}^2\right)} \end{aligned} $	$ \begin{aligned} & -\frac{2\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{3\epsilon} \\ & -\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \end{aligned} $	$ \begin{aligned} & -\frac{\hbar\log\left(\frac{\mu\ell^2}{M_{S1}^2}\right)\overline{W_{T1,T1}W_{T2,T2}}\ell_{T1}\ell_{a_4,T2}^TC P_{\ell_{a_2,T2}}H_{a_2}H_{a_3}M_{S1}^2\lambda_{T1}\ell_{a_1,T2}\ell_{a_3,T4}M_F^2}{2\left(M_F^2-M_{S1}^2\right)\left(M_F^2-M_{S2}^2\right)\left(M_{S1}^2-M_{S2}^2\right)} \\ & +\frac{\hbar\log\left(\frac{\mu\ell^2}{M_{S1}^2}\right)\overline{W_{T1,T1}W_{T2,T2}}\ell_{T1}\ell_{a_4,T2}^TC P_{\ell_{a$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
									$ \begin{aligned} & +C_{H2}H_{a_1}H_{a_1}^\dagger \\ & -H_{a_1}\mu^2H_{a_1}^\dagger \\ & +\frac{3hH_{a_1}H_{a_2}\lambda_{H1}^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}\lambda_{H1}^2H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{3hH_{a_1}H_{a_2}\lambda_{H2}^2H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{3}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}\lambda_{H2}^2H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & +\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & -\frac{\hbar\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & -\frac{\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \end{aligned} $	$ \begin{aligned} & -\frac{5\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{6\epsilon} \\ & -\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \end{aligned} $	$ \begin{aligned} & i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \\ & +i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \\ & +i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \\ & +i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \\ & +i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \\ & +i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \\ & +i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \\ & +i\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\frac{\overline{W_{1,1,1}W_{2,12,1}^*}\ell_{a_4,12}{}^TC P_L\ell_{a_2,11}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T1}\ell_{a_1,12}\ell_{a_3,14}M_F^2}{2\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_1}^2\right)\left(M_{S_1}^2-M_{S_1}^2\right)} \end{aligned} $
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$ \begin{aligned} & -\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon} \\ & -\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2 \\ & -\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2 \end{aligned} $			

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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